ADAPTIVE STUDENT PERFORMANCE ANALYZER

End-to-End System Architecture & Complete Operational Workflow Guide

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Target Audience: Administrators, HODs, Teachers, System Auditors

Platform Version: Student Marks Prediction v1.0 (Decision Tree ML & SHAP AI)

1. System Flow Summary Diagram

The diagram below illustrates the linear sequence of actions across Admin, HOD, Class Teacher, Student, and AI Automation layers:

Step / Phase	Operation Name	Technical Action & System Execution Description
STEP 01	**1. Admin**	Logs into System Admin Control Center (ADMIN001 / Rns@123)
STEP 02	**2. Creates HOD**	Creates Executive Department Account (e.g. HOD_CSE)
STEP 03	**3. Creates Class Teacher**	Creates Class Coordinator Account (e.g. CSE6A01)
STEP 04	**4. Credentials Generated**	System auto-generates Login ID & Temp Password (Rns@123)
STEP 05	**5. Credentials Export**	Admin downloads Credentials Report in Excel (.xlsx) & PDF
STEP 06	**6. Student Registers**	Student fills 5-section form with Dynamic Academic History
STEP 07	**7. Teacher Approves**	Teacher approves admission request in Pending Approval Queue
STEP 08	**8. Student Login**	Approved student logs into personal Dashboard
STEP 09	**9. Attendance Update**	Teacher records student class attendance percentage
STEP 10	**10. IA Mark Upload**	Teacher downloads protected Excel template & uploads IA-1/2/3 marks
STEP 11	**11. Automatic Calculation**	System calculates Best Two IA Avg, Attendance, CGPA & Backlogs
STEP 12	**12. AI Performance Prediction**	Decision Tree ML predicts Pass/Fail & Risk Level; SHAP attributes factors
STEP 13	**13. Study Plan Generation**	Personalized 7-Day Study Schedule generated for weak subjects
STEP 14	**14. Student Dashboard Sync**	Student Dashboard updates in real time with read-only controls
STEP 15	**15. PDF Report Download**	Teacher/Student downloads individual PDF & Complete Class PDF Report
STEP 16	**16. HOD Department Analytics**	HOD views executive analytics across Semesters 1 to 8
STEP 17	**17. Admin College Analytics**	Admin monitors college-wide statistics, user roles & system health

2. Detailed Step-by-Step Execution Guide

STEP 01	Admin Login & Access Control	Actor: Admin Role

The System Administrator logs in at http://localhost:5000/admin/login using ADMIN001 and password Rns@123. The Admin Control Center grants full college-wide governance privileges.

■■ **UI View:** Admin Login Form & System Administration Dashboard
• Displays total accounts, credentials export dropdown, and user creation controls.

STEP 02	Creating Head of Department (HOD) Accounts	Actor: Admin Role

Admin selects 'Head of Department (HOD)' in the account creation modal, assigns a department (e.g. CSE, ECE, ISE), and submits. The system auto-generates Login ID HOD_CSE.

■■ **UI View:** Account Creation Modal (Role: HOD, Dept: CSE)
• Auto-generates Login ID: HOD_CSE with default demo password Rns@123.

STEP 03	Creating Class Teacher Accounts	Actor: Admin Role

Admin creates a 'Class Teacher' account assigned to a specific class (e.g. CSE Sem 6 Sec A). The system auto-generates Login ID CSE6A01.

■■ **UI View:** Account Creation Modal (Role: Class Teacher, Dept: CSE, Sem: 6, Sec: A)
• Generates Login ID: CSE6A01 with default demo password Rns@123.

STEP 04	Teacher Credentials Auto-Generation & Security	Actor: System Automation

Every new account receives a unique User ID (e.g. USR-6307), Login ID (CSE6A01), and temporary password Rns@123 with first_login = 1 flag enforced for first-login password update.

■■ **UI View:** Success Alert & Hashed Password Storage Notice
• Passwords stored only as bcrypt/pbkdf2 hashes in database.

STEP 05	Admin Credentials Report Export (Excel & PDF)	Actor: Admin Role

Admin clicks 'Export Credentials Report' to download the master credentials file in **Excel (.xlsx)** or **Official PDF Document** containing User IDs, Full Names, Roles, Departments, Login IDs, and Temporary Passwords.

■■ **UI View:** Export Credentials Dropdown & Downloaded PDF/Excel Table
• Formats: Excel (.xlsx / .csv) & Confidential PDF Report.

STEP 06	Student Admission Registration & Academic History	Actor: Student Role

New students register at /student/register by completing 5 sections: Personal, Academic Background, Family, Home Study, and Technology. Selecting Current Semester = 6 automatically generates Semester 1 to 5 SGPA and backlog entry forms.

■■ **UI View:** Dynamic Academic History Form & Live Calculations Card
• Live calculation of Overall CGPA, Active Backlogs, Cleared Backlogs, and History.

STEP 07	Teacher Admission Approval Workflow	Actor: Class Teacher Role

Newly registered students start in 'Pending Approval' status. The Class Teacher logs in, reviews applicant details in the 'Pending Student Admission Approvals' queue, and clicks 'Approve'.

■■ **UI View:** Teacher Dashboard Pending Approval Table
• Approve / Reject action buttons; approved students gain dashboard login access.

STEP 08	Student Login & Session Protection	Actor: Student Role

Approved students log in at /student/login using USN or Email and Password. Session authorization restricts student access strictly to their own personal dashboard.

■■ **UI View:** Student Login Form & Welcome Banner
• Displays: 'Welcome, Hemanth' with department and USN subtitle.

STEP 09	Class Teacher Attendance Updating	Actor: Class Teacher Role

Teacher updates attendance percentages for class students from the Teacher Dashboard or via batch CSV/Excel sync. Attendance below 75% triggers attendance warning alerts.

■■ **UI View:** Teacher Roster Attendance Column & Summary Cards
• Class average attendance metric card automatically updates.

STEP 10	Structured Excel IA Mark Entry & Validation	Actor: Class Teacher Role

Teacher clicks 'Structured Excel Mark Entry', selects IA Test (IA-1, IA-2, IA-3), downloads pre-filled Excel template, enters marks (0-25/30 or 'AB'), and uploads. System validates row count, USNs, names, and ranges.

■■ **UI View:** Excel Template Generator Modal & Validation Success Card
• Rejects invalid ranges or altered student rows with clear error feedback.

STEP 11	Automatic System Calculations & Metrics Engine	Actor: System Automation

Upon mark entry, system automatically calculates: **Best Two IA Average** (selecting highest 2 of 3 IAs), overall **Attendance %**, **CGPA**, **Active Backlogs**, and **Cleared Backlogs**.

■■ **UI View:** Subject-Wise Ongoing IA Evaluation Table
• Columns: Subject Code, Subject Name, IA-1, IA-2, IA-3, Best Two IA Avg, Attendance.

STEP 12	AI Performance Prediction & SHAP AI Attributions	Actor: AI Engine (ML / SHAP)

Decision Tree ML models process student feature vectors to predict Result (Pass / Fail) and Risk Level (High, Medium, Low). SHAP explainable AI identifies top contributing factors.

■■ **UI View:** SHAP Explainable AI Attribution Card & Risk Badges
• Displays feature impact score (e.g. Attendance: +0.35, DBMS: +0.28).

STEP 13	Personalized AI Weekly Study Plan Generation	Actor: AI Engine (Planner)

Rules-based AI planner analyzes weak subjects (scored < 50%) and risk level to generate a custom 7-day study timetable (Monday to Sunday) specifying daily subjects and revision hours.

■■ **UI View:** Personalized Weekly Study Plan 7-Day Grid
• Breakdown from Monday to Sunday with target subjects and practice hours.

STEP 14	Student Dashboard Real-Time Synchronization	Actor: Student Role

The Student Dashboard updates in real time with 5 Summary Cards, IA Performance Table, SGPA Trend Graph, Teacher Remarks, Study Plan, Notifications, and Placement Readiness Index.

■■ **UI View:** Real-time Student Analytics Dashboard
• Fully read-only for students with strict security controls.

STEP 15	PDF Report Generation (Individual & Class PDF)	Actor: Teacher & Student

Clicking 'Download PDF Report' streams professional ReportLab PDF reports. Individual Student PDF contains complete academic analytics; Class PDF Report includes executive summary and student roster.

■■ **UI View:** Download PDF Report Buttons & Generated Document
• Official document styling with Coordinator and HOD signature blocks.

STEP 16	HOD Department Executive Analytics (Sem 1 to 8)	Actor: HOD Role

HOD logs in at /hod/login to view executive analytics for their department across all 8 semesters (Sem 1 to Sem 8). Filters student roster by semester and monitors high-risk trends.

■■ **UI View:** HOD Executive Portal & Semester Filter Buttons
• 6 summary cards, semester CGPA trends, and department PDF download.

STEP 17	Admin College-Wide System Oversight	Actor: Admin Role

Admin monitors college-wide statistics, edits faculty roles/class assignments, resets passwords, manages active/disabled accounts, and ensures seamless platform governance.

■■ **UI View:** Admin User Roster & Edit Role Modal
• Full system control across all departments, teachers, and students.